Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Heart

Date: 22 May 2025
Time: 12:24:09 PM

Female

Gene Symbols	Entrez Gene IDs	Probe IDs	BMD_{rd25} ($BMDL_{rd25}$ - $BMDU_{rd25}$) in ppm	Maximum Fold Change	Direction of Expression Change
Lck	313050	LCK_32705	31.7 (6.0 - NC)	1.4	DOWN
Spp1	25353	SPP1_9929	108.2 (30.6 - 345.5)	1.7	UP
Ccnb2	363088	CCNB2_8223	124.3 (51.4 - 277.6)	2.1	DOWN
Ccnb2-ps2	100364016	CCNB2_8223	124.3 (51.4 - 277.6)	2.1	DOWN
Ccna2	114494	CCNA2_8221	158.4 (60.3 - 308.2)	2.1	DOWN
Kif18b	303575	KIF18B_32882	165.4 (51.1 - 324.1)	2.3	DOWN
Prc1	308761	PRC1_9556	173.4 (45.9 - 395.4)	2.0	DOWN
Mki67	291234	MKI67_9232	176.8 (47.7 - 372.0)	1.9	DOWN
Rt1-da	294269	RT1-DA_9760	187.5 (72.3 - 411.3)	1.6	DOWN
Dbp	24309	DBP_8439	189.0 (183.6 - 228.8)	6.9	UP

rd25 = 25% relative deviation
NC = Non-convergent

Annotations:

Descriptions of orthologous human genes are shown due to the increased detail that is available in public resources such as UniprotKB (http://www.uniprot.org/uniprot/) and Entrez Gene (https://www.ncbi.nlm.nih.gov/gene/). Human UniprotKB was used as the primary resource due to the greater breadth of annotation and depth of functional detail that is provided. Rat UniprotKB was used as the second resource if the primary source did not provide a detailed description of function. Human Entrez gene summary was used as third resource. Rat Entrez gene summary was used as the fourth resource.

Gene definition version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

Lck: Human Uniprot function(Human LCK): Non-receptor tyrosine-protein kinase that plays an essential role in the selection and maturation of developing T-cells in the thymus and in the function of mature T-cells. Plays a key role in T-cell antigen receptor (TCR)-linked signal transduction pathways. Constitutively associated with the cytoplasmic portions of the CD4 and CD8 surface receptors. Association of the TCR with a peptide antigen-bound MHC complex facilitates the interaction of CD4 and CD8 with MHC class II and class I molecules, respectively, thereby recruiting the associated LCK protein to the vicinity of the TCR/CD3 complex. LCK then phosphorylates tyrosine residues within the immunoreceptor tyrosine-based activation motifs (ITAM) of the cytoplasmic tails of the TCR-gamma chains and CD3 subunits, initiating the TCR/CD3 signaling pathway. Once stimulated, the TCR recruits the tyrosine kinase ZAP70, that becomes phosphorylated and activated by LCK. Following this, a large number of signaling molecules are recruited, ultimately leading to lymphokine production. LCK also contributes to signaling by other receptor molecules. Associates directly with the cytoplasmic tail of CD2, which leads to hyperphosphorylation and activation of LCK. Also plays a role in the IL2 receptor-linked signaling pathway that controls the T-cell proliferative response. Binding of IL2 to its receptor results in increased activity of LCK. Is expressed at all stages of thymocyte development and is required for the regulation of maturation events that are governed by both pre-TCR and mature alpha beta TCR. Phosphorylates other substrates including RUNX3, PTK2B/PYK2, the microtubule-associated protein MAPT, RHOH or TYROBP. Interacts with FYB2 (PubMed:27335501). {ECO:0000269|PubMed:16339550, ECO:0000269|PubMed:16709819, ECO:0000269|PubMed:20028775, ECO:0000269|PubMed:20100835, ECO:0000269|PubMed:20851766, ECO:0000269|PubMed:21269457, ECO:0000269|PubMed:22080863, ECO:0000269|PubMed:27335501}.

Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley® SD®

Test Compound: 1,4-Dichlorobenzene (p-dichlorobenzene)
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Heart

Date: 22 May 2025
Time: 12:24:09 PM

Spp1: Human Uniprot function (Human SPP1): Binds tightly to hydroxyapatite. Appears to form an integral part of the mineralized matrix. Probably important to cell-matrix interaction.; FUNCTION: Acts as a cytokine involved in enhancing production of interferon-gamma and interleukin-12 and reducing production of interleukin-10 and is essential to the pathway that leads to type 1 immunity {ECO:000250}.

Ccnb2: Human Uniprot function (Human CCNB2): Essential for the control of the cell cycle at the G2/M (mitosis) transition.

Ccnb2-ps2: No description available.

Ccna2: Human Uniprot function (Human CCNA2): Cyclin which controls both the G1/S and the G2/M transition phases of the cell cycle. Functions through the formation of specific serine/threonine protein kinase holoenzyme complexes with the cyclin-dependent protein kinases CDK1 or CDK2. The cyclin subunit confers the substrate specificity of these complexes and differentially interacts with and activates CDK1 and CDK2 throughout the cell cycle. \{ECO:0000269;PubMed:1312467\}.

Kif18b: Human Uniprot function (Human KIF18B): In complex with KIF2C, constitutes the major microtubule plus-end depolymerizing activity in mitotic cells. Its major role may be to transport KIF2C and/or MAPRE1 along microtubules. \{ECO:0000269;PubMed:20600703, ECO:0000269;PubMed:21820309\}.

Prc1: Human Uniprot function(Human PRC1): Key regulator of cytokinesis that cross-links antiparrallel microtubules at an average distance of 35 nM. Essential for controlling the spatiotemporal formation of the midzone and successful cytokinesis. Required for KIF14 localization to the central spindle and midbody. Required to recruit PLK1 to the spindle. Stimulates PLK1 phosphorylation of RACGAP1 to allow recruitment of ECT2 to the central spindle. Acts as an oncogene for promoting bladder cancer cells proliferation, apoptosis inhibition and carcinogenic progression (PubMed:17409436). \{ECO:0000269;PubMed:12082078, ECO:0000269;PubMed:15297875, ECO:0000269;PubMed:15625105, ECO:0000269;PubMed:16431929,ECO:0000269;PubMed:17409436, 4;PubMed:19468300, ECO:0000269;PubMed:20691902, ECO:0000269;PubMed:9885575\}.

Mki67: Human Uniprot function (Human MKI67): Required to maintain individual mitotic chromosomes dispersed in the cytoplasm following nuclear envelope disassembly (PubMed:27362226). Associates with the surface of the mitotic chromosome, the perichromosomal layer, and covers a substantial fraction of the chromosome surface (PubMed:27362226). Prevents chromosomes from collapsing into a single chromatin mass by forming a steric and electrostatic charge barrier: the protein has a high net electrical charge and acts as a surfactant, dispersing chromosomes and enabling independent chromosome motility (PubMed:27362226). Binds DNA, with a preference for supercoiled DNA and AT-rich DNA (PubMed:10878551). Does not contribute to the internal structure of mitotic chromosomes (By similarity). May play a role in chromatin organization (PubMed:24867636). It is however unclear whether it plays a direct role in chromatin organization or whether it is an indirect consequence of its function in maintaining mitotic chromosomes dispersed (Probable). \{ECO:0000250&;UniProtKB:E9PVX6, ECO:0000269;PubMed:10878551, ECO:0000269; PubMed:24867636,ECO:0000269&;PubMed:27362226\}.

Rt1-da: No description available.

Dbp: Human Uniprot function (Human DBP): This transcriptional activator recognizes and binds to the sequence 5'-RTTAYGTAAY-3' found in the promoter of genes such as albumin, CYP2A4 and CYP2A5. It is not essential for circadian rhythm generation, but modulates important clock output genes. May be a direct target for regulation by the circadian pacemaker component clock. May affect circadian period and sleep regulation.

** END OF REPORT **